```
import pandas as pd
df = pd.read_csv('m2_metric_scenarios.csv')

df['my_accuracy']  = (df['true_positive']+df['true_negative'])/(df['true_negative'] + df['tru
df['my_precision'] = (df['true_positive']/(df['true_positive'] + df['false_positive']))
df['my_recall']    = (df['true_positive']/(df['true_positive'] + df['false_negative']))
df['my_f1']        = (2*(df['my_precision']*df['my_recall'])/(df['my_precision']+df['my_recal

df['total_error_cost'] = ((df['false_positive']*df['dollar_cost_per_false_positive']) + (df[

print(df[['scenario_id','accuracy','my_accuracy', 'precision', 'my_precision','recall','my_re
print('----------------------------------------------------------------------')
print('----------------------------------------------------------------------')
# check your work against the columns already in the file
print(df[['scenario_id','accuracy','my_accuracy', 'precision', 'my_precision','recall','my_re
print('----------------------------------------------------------------------')
print('----------------------------------------------------------------------')
print('S4 Sweep')
print('----------------------------------------------------------------------')
# a simple sweep. At each step you accept 4 more false positives
# in exchange for 1 fewer false negative.
fp, fn = 130, 95
cost_fp, cost_fn = 300, 3200
for step in range(0, 6):
    f_p = fp + step
    f_n = fn - step
    total = (f_p*cost_fp) + (f_n * cost_fn)
    print(step, f_p, f_n, total)
print('----------------------------------------------------------------------')
```

```
  scenario_id  accuracy  my_accuracy  precision  my_precision  recall  \
0          S1    0.9900     0.990000     0.8333      0.833333  0.9524
1          S2    0.9918     0.991836     0.9266      0.926641  0.7947
2          S3    0.9930     0.992986     0.8757      0.875706  0.9226
3          S4    0.9536     0.953560     0.8000      0.800000  0.8455

   my_recall      f1     my_f1  total_error_cost
0   0.952381  0.8889  0.888889            500008
1   0.794702  0.8556  0.855615              7520
2   0.922619  0.8986  0.898551             27140
3   0.845528  0.8221  0.822134            343000
----------------------------------------------------------------------
----------------------------------------------------------------------
  scenario_id  accuracy  my_accuracy  precision  my_precision  recall  \
0          S1    0.9900     0.990000     0.8333      0.833333  0.9524
3          S4    0.9536     0.953560     0.8000      0.800000  0.8455
2          S3    0.9930     0.992986     0.8757      0.875706  0.9226
1          S2    0.9918     0.991836     0.9266      0.926641  0.7947

   my_recall      f1     my_f1  total_error_cost
0   0.952381  0.8889  0.888889            500008
3   0.845528  0.8221  0.822134            343000
2   0.922619  0.8986  0.898551             27140
1   0.794702  0.8556  0.855615              7520
----------------------------------------------------------------------
----------------------------------------------------------------------
S4 Sweep
```

```
-----------------------------------------------------------------
0 130 95 343000
1 131 94 340100
2 132 93 337200
3 133 92 334300
4 134 91 331400
5 135 90 328500
-----------------------------------------------------------------
```